MIL-H-43224G
18 December 1981
SUPERSEDING
MIL-H-43224F
22 May 1979

# MILITARY SPECIFICATION

## HASH, BEEF, DEHYDRATED, COOKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements for cooked, dehydrated, beef hash for use by the Department of Defense as a component of operational rations.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| L-P-378 | - | Plastic Sheet and Strip, Thin Gauge, Polyolefin |
| EE-B-575 | - | Bouillon (Soup and Gravy Bases) |
| EE-S-631 | - | Spices, Ground and Whole, and Spice Blends |
| QQ-A-1876 | - | Aluminum Foil |
| JJJ-S-791 | - | Sugar, Refined and Brown, Beet or Cane |
| PPP-B-26 | - | Bags, Plastic (General Purpose) |
| PPP-C-843 | - | Cushioning Material Cellulosic |
| PPP-C-1797 | - | Cushioning Material, Resilient, Low Density, Unicellular Polypropylene Foam |
| PPP-T-60 | - | Tape Packaging, Waterproof |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8940

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MILITARY

MIL-L-35078 - Loads Unit, Preparation of Nonperishable Subsistence Items; General Specification For

STANDARDS

FEDERAL

FED-STD-101 - Preservation, Packaging and Packing Materials: Test Procedures
FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-668 - Sanitary Standards for Food Plants

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)



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THE UNITED STATES PHARMACOPOEIAL CONVENTION, INC.

Pharmacopoeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, 20th and Northhampton Streets, Easton, PA 18042.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D 882 - Tensile Properties of Thin Plastic Sheeting

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Suite 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, Suite 1900, 601 California Street, San Francisco, CA 94108.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

## 3. REQUIREMENTS

3.1 First article. When specified (see 6.1), the contractor shall furnish a sample for first article inspection and approval (see 4.4 and 6.3).



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3.2 Material. All material shall be in excellent condition, clean, sound, wholesome, and free from evidence of insect and rodent infestation, foreign and undesirable odors, flavors, colors, and extraneous materials. It shall be so handled and processed that the end product shall comply with the finished product requirements of 3.6.

3.2.1 Beef. The beef shall be from steers, heifers, or cows and shall be derived from any combination of carcasses, quarters, and recognizable primal and sub-primal cuts (square-cut chuck, ribs, trimmed full loins, trimmed shortloins, loin ends, striploins, rounds (heel (gastrocnemius muscle) shall be excluded), shoulder clods, chuck rolls, ribeyes, top rounds, knuckles, and bottom rounds ,shoulder clods, chuck rolls, ribeyes, top rounds, knuckles, and bottom rounds). Recognizable cuts are those which, when compared to IMPS cuts, have no more than a minor amount of lean, fat, or bone removed or included from an adjacent cut. Rough cuts (foreshanks, flanks, short plates, briskets, rib covers, necks, hindshanks, etc.) shall be excluded. All suitable lean meat shall be used except tenderloins may be excluded at the option of the contractor. The beef shall be in the fresh chilled state and shall be in excellent condition, i.e., exposed lean and fat surfaces shall be of a color and bloom normally associated with the class and cut of meat, and typical of meat which has been properly stored and handled; cut surfaces and naturally exposed lean surfaces shall show no discoloration; the fat shall show no discoloration; no odors foreign to fresh meat shall be present; and there shall be no evidence of freezing, defrosting, or mishandling.

3.2.1.1 Boning and trimming. The beef shall be boned and trimmed to remove objectionable material such as bone, cartilage, heavy connective tissue, etc. The boneless trimmed meat shall meet the limitations specified in tables I and II.

3.2.1.2 Holding temperature and time of boneless frozen beef. The boneless beef may be frozen and held for not more than 90 days at a temperature of 0°F or lower from time of initial freezing until start of processing.

3.2.2 Beef fat. Fat shall be fat derived from trimming authorized cuts or cod fat having the characteristics of cod fat from U.S. Grade Good or higher grade beef. It shall be in excellent condition at the time of use and shall be free from rancidity, sourness and foreign odor. It shall be white to light yellow in color. The fat used in the product shall be free of foreign materials or any other extraneous materials.

3.2.3 Garlic powder. Garlic powder shall comply with the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products for Standard grade or better garlic powder.

3.2.4 Onion powder. Onion powder shall comply with the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products for Standard grade or better onion powder.



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3.2.5 Potatoes. Potatoes shall be one of the following varieties: Russet Burbank, Kennebec, Green Mountain, Cobbler, Red McClure, Sebago, Katahdin, or Snowflake. Dehydrated or dehydro-frozen potatoes shall not be used.

3.2.6 Salt. Salt shall be white, refined sodium chloride with or without anticaking agent. Iodized salt shall not be used.

3.2.7 Soup and gravy base, beef flavored. Beef flavored soup and gravy base shall comply with type I of EE-B-575.

3.2.8 White pepper. White pepper shall comply with type II of EE-S-631.

3.2.9 Sugar. Sugar shall comply with type I, class (a) of JJJ-S-791.

3.2.10 Nitrogen. Nitrogen shall meet the requirements of the Pharmacopoeia of the United States and shall be water or liquid nitrogen pumped.

3.3 Processing.

3.3.1 Grinding of beef. Fresh or frozen, boned and trimmed cuts of beef, as specified in 3.2.1, shall be ground twice as described hereinafter. If it is necessary to add fat, it shall be mixed in before, during or just after the coarse grind. The beef shall be coarse ground through a plate having holes 3/4 to 1-1/2 inches in diameter. It shall then be ground through a plate having holes 3/8 inch in diameter. The temperature of fresh beef shall not exceed 50°F on completion of either grinding operation. Fresh ground beef shall be held at 40°F or lower for not longer than 72 hours between grinding of beef and start of processing. Alternatively fresh ground beef may be frozen after either grind as specified in 3.3.1.1 or 3.3.1.2. Frozen beef may be tempered to not higher than 30°F to facilitate grinding and shall not be considered to have been thawed if it is placed in a freezer at 0°F or lower immediately after grinding. Frozen meat that has been thawed shall not be refrozen prior to freezing of the finished product for dehydration.

3.3.1.1 Frozen coarse ground beef. If fresh ground beef as specified in 3.3.1 is frozen after the coarse grind, it shall be packaged to protect product from contamination or freezer burn and frozen to an internal temperature of 0°F or lower within 72 hours after grinding. The frozen coarse ground beef shall be held at 0°F or lower for not longer than 75 days from the time of grinding to the start of processing.

3.3.1.2 Frozen ground beef. If fresh ground beef as specified in 3.3.1 is frozen after the final grind, it shall be packaged to protect the product from contamination or freezer burn and frozen to an internal product temperature of 0°F or lower within 72 hours after grinding. The frozen ground beef shall be held at 0°F or lower for not longer than 20 days from the time of grinding to the start of processing.

3.3.1.3 Fat content. Fat content of the ground beef shall be in the range of 14 to 21 percent when determined by the Thermal Extraction Method or 14 to 23 percent when determined by the Soxhlet Method. Testing shall be as specified in 4.5.3.2.



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3.3.2 Potato processing. The potatoes shall be washed, sorted, peeled, trimmed to remove defects, and mechanically diced with dicer settings at approximately 3/8 by 3/8 by 3/8-inch. They shall be processed according to good commercial practice.

3.3.3 Formulation. The beef hash shall be prepared in accordance with the following formula:

| | Percent by weight |
|---|---|
| Beef, raw, ground | 43.25 |
| Potatoes, diced, raw | 40.25 |
| Seasoning mix | 4.11 |
| Water | 12.39 |
| | 100.00 |

| Seasoning mix | Percent by weight |
|---|---|
| Onion powder | 48.66 |
| Salt | 24.33 |
| Soup and gravy base | 15.80 |
| Sugar | 6.10 |
| Garlic powder | 3.65 |
| Pepper, white, ground | 1.46 |
| | 100.00 |

3.3.4 Cooking. The potatoes shall be cooked in water until tender but not mushy and rinsed with cold water. Seasoning mix and water shall be stirred while heating to 180°F. The ground meat shall be added and the mixture cooked until the red color of the meat disappears and the mixture reaches a temperature of at least 180°F. The cooked potatoes shall be added and the mixture again heated to at least 180°F. Any process which will produce an equivalent product shall be acceptable. If it is found that less water than specified in 3.3.3 can be used, the quantity of water may be decreased. However, increased amounts of water shall not be used unless the density and moisture absorption properties of the raw materials required additional water to accomplish the desired effect. The prepared product shall be spread on dehydrator trays and frozen.

3.4 Freeze dehydration. The product shall be freeze-dehydrated at an absolute pressure not to exceed 1.5 millimeters of mercury and a product temperature as indicated by suitable instruments, not to exceed 150°F. If the platen temperature is maintained at 155°F or below with radiant heating, the product temperature may be disregarded. After dehydration is completed, the pressure shall be equalized to atmospheric level with nitrogen and the product shall be preserved immediately as specified in 5.1. If it is necessary to hold the product more than 2 hours between dehydration and preservation, it shall be adequately protected from oxygen and moisture by either holding under a nitrogen atmosphere with 2.0 percent or less oxygen, or under a vacuum of at least 27 inches of mercury. If vacuum is used it shall be broken with nitrogen.



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3.5 Time and temperature limitations. The cooked beef hash shall be so handled as to comply with the following limitations:

| | |
|---|---|
| (1) Maximum holding time of cooked beef hash before freezing. | 24 hours |
| (2) Maximum internal product temperature of cooked beef hash before freezing 1/ | 40°F |
| (3) Maximum holding time of frozen cooked beef hash before being placed in freeze dehydration | 5 days |
| (4) Maximum temperature of frozen cooked beef hash prior to freeze dehydration | 0°F |

1/ Temperature specified is the temperature of the product itself. To prevent undue holding of the cooked product above 40°F the product temperature shall be dropped to 40°F or lower within a period of 3 hours.

3.6 Finished product.

3.6.1 Physical characteristics. The finished product shall comply with the following physical characteristics:

Dehydrated product:

(1) There shall be no foreign material which is extraneous to the product such as but not limited to glass, paint, wrapping paper or casing material.

(2) There shall be no evidence of incomplete dehydration such as damp or soggy areas.

(3) There shall be no evidence of thawing in the dehydrator such as glazed surface areas measuring more than 1/2-inch in any dimension or dark colored cores of any dimension.

Rehydrated product;

(4) When rehydrated in hot water 180° to 200°F using the proportions of 13 ounces of water to 5 ounces of beef hash for 10 minutes, the product shall show complete water penetration through all particles. However, gristle and similar material in the meat that do not necessarily rehydrate completely shall not be considered evidence of incomplete rehydration.

(5) The rehydrated product shall possess flavor, odor, and texture considered normal for frozen and heated beef hash. There shall be no scorched, burnt, or foreign flavors, odors, or colors.

(6) When rehydrated as in (4) above, the product shall show distinct pieces of potato and distinguishable particles of meat such as will be found in cooked ground meat. It shall not be similar to mashed potatoes with ground meat.



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3.6.2 Analytical requirements. The finished product shall contain a maximum of 2.0 percent moisture.

3.6.3 Microbiological requirements. The E. coli count shall be less than 3 per gram (no positive tube in the standard 3-tube MPN technique) in at least four of five samples. One of the five samples from one specific lot of the product may have an MPN up to but not greater than 10 per gram. The total aerobic plate count shall not exceed 75,000 per gram.

3.7 Plant qualification. The product shall originate, be processed, marked and handled in plants which are operated under the Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture. Finished product refrigeration facilities (freezer, storage and shipping) not operating under these regulations shall meet the requirement of MIL-STD-668.

3.8 Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Plant qualification conditions. The product shall be unacceptable if produced and stored in plants which do not meet the requirements specified in 3.7.

4.4 First article inspection. When a first article is required (see 6.1) it shall be examined for the defects specified in table IX and tested for the characteristics specified in 4.5.5. The presence of any defect or failure of any test shall be cause for rejection of the first acticle.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.



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4.5.1.1 Beef examination. All beef shall be examined in either the bone-in or boneless state for conformance to the condition and cut requirements in 3.2.1. Cuts initially examined in the boneless state shall be in the form of whole boneless recognizable cuts. Any nonconforming beef shall be rejected.

4.5.1.2 Boned and trimmed beef examination. The beef shall be examined after boning and trimming and prior to any further processing or to any freezing. The lot size expresssed in terms of pounds shall be as declared to the AMS agent by the contractor. However, the AMS agent reserves the right to declare as a lot a portion of a declared lot, if, in his or her opinion, that portion may be out of compliance with any of the specification requirements. The sampling plans for boneless product are listed in table I. The sample unit shall be a minimum of 12 pounds of adjacent boneless beef. If all or a portion of the sample falls within a larger cut, the entire cut(s) shall be examined. The product shall be examined for defects listed in table II. Failure of the product to meet the acceptance criteria as identified in table I shall cause rejection of the involved product. Except for product rejected because of evidence of freezing or defrosting or presence of any off-condition, or if it has been slashed or ground, it may be reworked by the contractor and re-offered for examination. For re-examination the sampling plan used shall be the one in table I designed for the next larger lot size than the one under which the lot was initially rejected. Product shall not be re-examined more than one time.

TABLE I. Sampling plan for boned and trimmed beef

| Lot size (pounds) | Sample size (No. sample units) | Defect categories | | | |
|---|---|---|---|---|---|
| | | Major | | Minor | |
| | | AC | RE | AC | RE |
| 500 or less | 13 | 1 | 2 | 3 | 4 |
| 501 to 1200 | 20 | 2 | 3 | 5 | 6 |
| 1201 to 3200 | 32 | 3 | 4 | 7 | 8 |
| 3201 to 10000 | 50 | 5 | 6 | 10 | 11 |
| 10001 to 35000 | 80 | 7 | 8 | 14 | 15 |
| Reinspection lots of 10001 to 35000 | 125 | 10 | 11 | 21 | 22 |



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TABLE II. Boned and trimmed beef defects

| Category | | Defect 1/ 2/ |
|---|---|---|
| Major | Minor | |
| 151 | | Presence of lymph gland measuring 1.0 inch or more in any dimension. |
| 152 | | Presence of blood clot measuring 0.5 inch or more in any dimension. |
| 153 | | Presence of bruise measuring 1.0 inch or more in any dimension. |
| 154 | | Presence of bone measuring 0.3 inch or more in any dimension. |
| 155 | | Presence of cartilage measuring 0.5 inch or more in any dimension. |
| 156 | | Presence of backstrap or other ligament measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other). |
| 157 | | Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 2.0 square inches or more. |
| 158 | | Presence of knuckle cover. |
| 159 | | Presence of kidney, pizzle eye, pre-pubic tendon, thymus gland or hanging tender measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other). |
| 160 | | Presence of heel (gastrocnemius muscle). |
| | 201 | Presence of lymph gland measuring more than 0.5 inch but less than 1.0 inch in any dimension. |
| | 202 | Presence of blood clot measuring 0.3 inch or more but less than 0.5 inch in any dimension. |



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TABLE II. Boned and trimmed beef defects (cont'd)

| Category | | Defect 1/ 2/ |
|---|---|---|
| Major | Minor | |
| | 203 | Presence of bruise measuring 0.5 inch or more but less than 1.0 inch in any dimension. |
| | 204 | Presence of shank, clod, knuckle, or bottom (outside) round with tendinous end showing less than 75 percent lean tissue on a cross-sectional cut surface. |
| | 205 | Presence of backstrap or other ligament measuring 0.5 inch or more but less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other). |
| | 206 | Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 0.5 square inches or more but less than 2.0 square inches. |
| | 207 | Presence of dehydrated surface measuring 1.0 square inch or more. |
| | 208 | Presence of discolored meat (including blood discolored neck meat) measuring 1.0 square inch or more. |
| | 209 | Presence of exposed major blood vessel measuring 1.0 inch or more in any dimension. |
| | 210 | Presence of calcified (scratchy) periosteum measuring 2.0 square inches or more. |
| | 211 | Presence of kidney, pizzle eye, pre-pubic tendon, hanging tender or thymus gland measuring less than 1.0 inch in longest dimension. |
| | 212 | Presence of heavy connective tissue on lower edge of short plate or flank. |
| | 213 | Presence of membranous portion of diaphragm or membranous covering from skirt, flank or abdominal section of short plate measuring 3 square inches or more. |



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TABLE II. Boned and trimmed beef defects (cont'd)

| Category | | Defect 1/ 2/ |
|---|---|---|
| Major | Minor | |
| | 214 | Presence of heavy connective (abdominal tunic) tissue on the flank measuring 1.0 inch or more in any dimension. |
| | 215 | Presence of cod, udder, kidney or pelvic fat. |

1/ Acceptability of product with respect to the presence of foreign material (for example: glass, dirt, insect parts, hair, wood, metal) shall be made by a Meat and Poultry Inspection Program employee. The presence of any product not in excellent condition shall cause rejection of the involved lot.

2/ Evidence of freezing or defrosting or presence of any off-condition -- including but not restricted to being off-odor, slightly sticky, gassy, rancid, or sour, or showing discoloration -- shall cause rejection of the involved lot.

4.5.1.3 Beef fat examination (when applicable). When addition of beef fat is necessary (see 3.3.1 and 3.3.1.3), the fat shall be examined for the defects listed in table III prior to grinding. The lot size unit and the sample unit shall be 1 pound of beef fat. The sample size shall be the number of sample units indicated by inspection level I. The presence of any defect shall be cause for rejection of the lot of beef fat or the finished product made therefrom, as applicable.

TABLE III. Beef fat defects

| Category | Defect |
|---|---|
| Major A | |
| 101 | Fat not derived from trimmings of authorized cuts or not having the characteristics of cod fat derived from U.S. Grade Good or better grade beef. |
| 102 | Not white to light yellow in color. |
| 103 | Not in excellent condition (e.g., moldy, slimy, sticky, watery, etc.) |
| 104 | Presence of rancid, sour or foreign odor. |
| 105 | Presence of foreign materials (e.g., wood, paper, glass, metal or any other extraneous material). |



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4.5.1.4 Examination of garlic powder and onion powder. Examination shall be made to determine compliance with the identity requirements of 3.2.3 and 3.2.4. Determination of compliance shall be ascertained by examination of labels, invoices or other valid documents. Nonconformance to the above referenced requirements shall be cause for rejection of the component product or the finished product made therefrom, as applicable.

4.5.1.5 Examination of soup and gravy base (beef flavored) and white pepper. The above mentioned ingredients shall be examined in accordance with the examination criteria of the applicable specification for product characteristics referenced in th pertinent component paragraphs, except that the sampling plans and AQLs shall not apply. The sample for examination shall be a 4-ounce composite for each ingredient, derived from five primary containers or all containers if less than five form a lot. Use of ingredients having one or more defects shall be cause for rejection of the involved quantity of finished product made therefrom.

4.5.1.6 Examination of potatoes and salt. Examination of these components to determine compliance with the requirements specified in 3.2.5 and 3.2.6 as concerns identity (variety, grade, type), shall be ascertained by examination of labels, invoices, or other valid documents. In addition, these items shall be examined for the condition requirements specified in 3.2. The sample for examination for potatoes shall be a 5-pound composite and for salt shall be a 1/2-pound composite derived from five primary containers or all containers if less than five form a lot. Finding of one or more defects shall indicate an unacceptable ingredient and shall be cause for rejection of the component material or any finished product made therefrom.

4.5.1.7 Examination of nitrogen. Examination shall be made to determine compliance with the identity requirements of 3.2.10. Determination of compliance shall be ascertained by examination of labels, invoices or other valid documents. Nonconformance to the above referenced requirements shall be cause for rejection of the component product or the finished product made therefrom, as applicable.

4.5.2 Component testing. In addition to any testing required by 4.5.1, component shall be tested as indicated in table IV. Results shall be applicable to the lot average, except for laminated plastic-foil bag material which shall be both individu specimen and unit average requirements. Nonconformance to one or more test requirements shall be cause for rejection of the lot.



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TABLE IV. Component test (see 4.5.1)

| Component | Sample unit | Lot size expressed in | Inspection level | Characteristic | Results reported to nearest | Test ref and rqmt. paragraph |
|---|---|---|---|---|---|---|
| Polyethylene bag | Bag | Bags | S-1 | Thickness | 0.0001 inch | 5.1.1<br>4.6.1 |
| | | | | Seal strength | One-quarter pound | 5.1.1<br>4.6.6 |
| Laminated 1/ plastic foil bag material | Bag or sheet | Bags or sheets | S-1 | | | |
| Ionomer or polyethylene film | | | | Thickness | 0.0001 inch | 5.1.2<br>4.6.1 |
| Aluminum foil | | | | Thickness | 0.00001 inch | 5.1.2<br>4.6.1 |
| Polyester or Nylon 6 | | | | Thickness | 0.0001 inch | 5.1.2<br>4.6.1 |
| Polyethylene | | | | Weight | One pound | 5.1.2<br>4.6.1 |
| Laminated foil bag 2/ | | | | Seal strength | One-quarter pound | 5.1.2<br>5.1.2.1<br>4.6.6 |
| Baseboard | Board | Boards | S-1 | Thickness | 0.001 inch | 5.1.1.2<br>4.6.2 |

1/ In lieu of testing the thickness of each lamina, determination of compliance for thickness may be ascertained by the examination of labels, invoices or other valid documents.

2/ Separate samples of empty component bags and sealed bags of the end item shall be submitted for testing bag manufacturer's seams and the end product producer's seams.



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4.5.3 In-process inspection.

4.5.3.1 In-process examination. Examination shall be made during processing to determine compliance with requirements of 3.2.1.2, 3.3 through 3.5 and 5.1.2.1 as concerns preparation, formulation, time and time intervals, temperatures, absolute pressure, use of nitrogen or vacuum and other applicable procedures. Time, temperature, pressure and formulation records shall be maintained. Noncompliance with one or more requirements either by records or reflected by actual examination shall be cause for rejection of the lot or the finished product made therefrom, as applicable.

4.5.3.2 In-process testing for fat content. The ground beef shall be tested for fat content as specified in 4.6.5.1 or 4.6.5.2. The lot size shall be expressed in pounds. The sample shall be a 1/2-pound composite derived from the number of 1-pound units indicated by inspection level S-2. The result shall be reported to the nearest 1 percent. Failure to conform to the fat requirement in 3.3.1.3 shall be cause for rejection of the lot.

4.5.4 End item and packaging examination. Classification of defects found during examination shall be in accordance with tables VI through X. The examination criteria for the above mentioned tables shall be as shown in table V. The AQL for table VII shall be expressed as percent defective and in terms of defects per hundred units for tables VI, VIII, IX and X.

TABLE V. End item and packaging examination criteria

| Table | Inspection level | Sample unit | Lot size expressed in | AQLs | | |
|---|---|---|---|---|---|---|
| | | | | Major A | Major B | Minor |
| VI | S-2 | One unfilled bag | Bags | - | 1.5 | 6.5 |
| VII | S-3 | One filled and sealed bag | Bags | - | - | 2.5 |
| VIII | S-3 | One filled and sealed bag | Bags | - | 1.5 | 6.5 |
| IX | S-2 | Contents of one filled and sealed bag | Bags | 1/ | 1.5 | - |
| X | S-3 | One shipping container | Shipping containers | - | 1.5 | 6.5 |

1/ Presence of one or more major A defects shall be cause for rejection of the lot.



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TABLE VI. Bag (and baseboard when applicable) defects

| Category | | Defect | | |
|---|---|---|---|---|
| Major B | Minor | (Applicable prior to filling and sealing) | Polyethylene bag | Laminated plastic-foil bag |
| 151 | | Dimensions not as specified (see 5.1.1 and 5.1.2) 1/ | X | X |
| 152 | | Not style or material specified | X | X |
| 153 | | Three edges not heat sealed or width of heat seals less than 1/4 inch or more than 1/2 inch | | X |
| | 201 | Lip does not extend 1/4 inch ± 1/16 inch beyond outer edge (when applicable) 2/ | | X |
| | 202 | "V" or "C" shaped tear notch less than 1/32-inch deep or distance between inside edge of notch and inside edge of seal less than 3/16 inch 2/ | | X |
| | 203 | "V"or "C" shaped notch not located between 1-1/4 to 1-1/2 inch from top edge of bag or not in both side seals 2/ | | X |
| 154 | | Color not as specified (see 5.1.2) | | X |
| 155 | | Evidence of delamination | | X |



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TABLE VI. Bag (and baseboard when applicable) defects (cont'd)

| Category | | Defect | | |
|---|---|---|---|---|
| Major B | Minor | (Applicable prior to filling and sealing) | Polyethylene bag | Laminated plastic-foil bag |
| | 204 | Tape missing or not placed over ears (when applicable, see 5.1.1) | X | |
| | | (Applicable to baseboard) | | |
| | 205 | Baseboard material not as specified | | |
| | 206 | Not a natural color ranging from brown to gray when applicable | | |
| | 207 | Not dimensions specified | | |
| | 208 | Corners not rounded | | |

1/ Bag dimensions are inside dimensions exclusive of seams or seals. Deviations from bag dimensions due to seal creep or seams not being parallel with each other shall be scored as a defect.

2/ Measure to nearest 1/16 inch or 1/64 inch as applicable to the basic dimensions.

TABLE VII. Net weight defect 1/

| Category | Defect |
|---|---|
| Minor | |
| 201 | Net weight less than 4-3/4 ounces |

1/ The lot shall be rejected if the average net weight of the samples is less than the net weight specified. Weigh to the nearest 1/8 ounce.



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TABLE VIII. Filled and sealed bag defects

| Category | | Defect | | |
|---|---|---|---|---|
| Major B | Minor | | Polyethylene bag | Laminated foil bag |
| 151 | | Not formed as specified | X | X |
| 152 | | Broken, sifter, tear or hole | X | X |
| | 201 | Maximum thickness exceeds 2-1/4 inches at any point 1/ | X | |
| | 202 | Not folded as specified | X | |
| 153 | | Not clean | X | X |
| 154 | | Name of product missing, incorrect or illegible | | X |
| | 203 | Other markings missing, incorrect or illegible | | X |
| 155 | | Color of label white or other than dull non-reflecting color | | X |
| 156 | | Printing not bold black face or indelible ink 2/ | | X |
| 157 | | Closure seal less than 1/4 inch or more than 1/2 inch in width or the seal width is less than 5/16 inch for double closure seal or less than 1/8 inch for thermal impulse seal 3/ | | X |
| | 204 | Label not affixed as specified or alternative printing not as specified | | X |



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TABLE VIII. Filled and sealed bag defects (cont'd)

| Category | | Defect | | |
|---|---|---|---|---|
| Major B | Minor | | Polyethylene bag | Laminated foil bag |
| | 205 | Tear notch not below closure seal or 1/16 inch thick, as applicable | | X |
| 158 | | Material imparts off-odor or flavor to product | X | X |

1/ Measure to nearest 1/4 inch.

2/ When applicable, determination of compliance to the indelible ink requirements shall be ascertained by examination of labels, invoices or other valid documents.

3/ Measure to nearest 1/8 inch, 1/16 inch or 1/32 inch as applicable to the basic dimension.

TABLE IX. Product defects

| Category | | Defect |
|---|---|---|
| Major A | Major B | |
| | | Applicable to dehydrated product |
| 101 | | Presence of foreign material (e.g., glass, paint, etc.) or presence of foreign material which is extraneous to the product, but not limited to wrapping paper or casing material. |
| | 151 | Evidence of incomplete dehydration, e.g., damp or soggy areas. |
| | 152 | Glazed surface areas measuring more than 1/2 inch in any dimension. 1/ |
| | 153 | Presence of dark colored cores of any dimension. |



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TABLE IX. Product defects (cont'd)

| Category | | Defect |
|---|---|---|
| Major A | Major B | |
| | | Applicable to rehydrated product 2/ |
| 102 | | Presence of scorched, burnt, or foreign flavors, odors or colors. |
| | 154 | Incomplete water penetration through all particles. 3/ |
| | 155 | Flavor, odor, and texture not considered normal for frozen and heated beef hash or not comparable to first article sample. |
| | 156 | Product does not show distinct pieces of potato and distinguishable particles of meat (i.e., in contrast to mashed potatoes with ground meat). |

1/ Measure to nearest 1/8 inch.

2/ See item (4) of 3.6.1 for rehydration procedure.

3/ Dry or callous-like areas attributable to gristle and similar material shall not be considered as defects.

TABLE X. Packing defects

| Category | | Defect |
|---|---|---|
| Major B | Minor | |
| 151 | | Fiberboard box or liner or top and bottom pads and dunnage not material specified (see 5.2.1). |
| 152 | | Top or bottom or both pads missing. |
| | 201 | Arrangement not as specified (see 5.2.1). |
| | 202 | Package material not in flat (unfolded) state. |
| | 203 | Dunnage material not placed as specified (see 5.2.1). |
| | 204 | Length and width of dunnage material less than the length and width of shipping container. 1/ |



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TABLE X. Packing defects (cont'd)

| Category | | Defect |
|---|---|---|
| Major B | Minor | |
| | 205 | Thickness of dunnage material less than 1/8-inch thick or 1/16 inch thick, as applicable. 2/ |
| 153 | | National stock number, item description, contract number or date of pack markings missing, incorrect, or illegible. |
| | 206 | Other required markings missing, incorrect, or illegible. |
| | 207 | Less than 45 filled and sealed packages. 3/ |

1/ Measure to nearest 1 inch.

2/ Measure to nearest 1/16 inch for 1/8 inch thick material and to nearest 1/32 inch for 1/16 inch thick material.

3/ Lot shall be rejected if the average count is less than the specified count.

4.5.4.1 Examination for loss of vacuum in flexible food packet. The evacuated packets shall be examained on a 100 percent basis to determine compliance with the vacuum requirements of 5.1.2.1. The examination shall be made not less than 24 hours after sealing. During the examination the package shall continue to exhibit a tight adherence of the bag to the surface contours of the contents along the seams when a force is applied away from the contents at the center of each side seam (holding each side seam between thumb and forefinger of each hand, simultaneously exert a slight pull with both hands). Repackaging of the product into a new laminated foil plastic bag will be permitted provided that the repackaging and resealing is accomplished within 7 days of the original sealing. Any bag showing a vacuum failure shall be rejected. Records shall be maintained to reflect the percent examination and the results thereof.

4.5.4.2 Examination of unit loads. When applicable, examination of unit loads shall be in accordance with MIL-L-35078 and the applicable specification sheet for the type and class of load specified.

4.5.5 End item testing. The finished product (dehydrated form) shall be tested as specified in 4.6.3 and 4.6.4 for conformance to the moisture and microbiological requirements in 3.6.2 and 3.6.3. For determination of moisture, the sample shall be one filled and sealed primary container, the lot size shall be expressed in terms of primary containers, the inspection level shall be S-2, and the AQL shall be 1.5 expressed as percent defective. Any failure of a moisture test shall be considered a major defect. For the microbiological determination, the sample unit shall be one primary container, the sample size shall be five sample units and any nonconformance to the requirements of 3.6.3 shall be cause for rejection of the lot.



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4.6 Methods of inspection.

4.6.1 Test procedures for determining laminae thickness. These procedures are to apply to tests that are to be conducted at the film laminator's plant to verify the following material structure:

Laminated polyethylene
Aluminum foil
Extruded polyethylene
Polyester or Nylon 6

4.6.1.1 Polyethylene lamina. The thickness of the polyethylene lamina shall be tested in accordance with L-P-378, except that a machinist's micrometer may be used, provided graduations and accuracy conform to the requirements stated in L-P-378.

4.6.1.2 Aluminum foil. The thickness of the foil shall be determined in accordance with the procedure specified in QQ-A-1876.

4.6.1.3 Polyethylene extrusion.

(a) If the equipment performing the extrusion operation is equipped with a beta gauge reading device, direct readings shall be observed and these will be regarded as the thickness of the extrusion. Alternatively, the following method can be utilized:

(b) A sample of at least 4 by 4 inches shall be cut from the completed material immediately after the extrusion operation is completed on the specific roll being constructed. Separation of the laminae shall be attempted, under proper conditions, utilizing a solvent such as tetrahydrofuran or toluene (separation of the laminae can be accomplished easier immediately after the material has been fabricated before the adhesives have cured). If this method is successful, the extrusion coating shall be removed, dried, and tested for thickness as specified for the polyethylene lamina. NOTE: Of the two methods specified, the first is considered to be the more accurate.

4.6.1.4 Polyester or Nylon 6. The thickness of the polyester or Nylon 6 lamina shall be determined by a gauge measurement as specified for the polyethylene lamina.

4.6.2 Thickness of baseboard. Thickness of baseboard shall be determined in accordance with Method 1003 of FED-STD-101.

4.6.3 Moisture test. Moisture in the product shall be determined in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists; Chapter: Meat and Meat Products; Section: Meat; Method: Drying in Vacuum, except that the temperature time cycle shall be modified by using a temperature of 70°C for a period of 16 hours at a pressure of 13 kPa. Moisture shall be reported to the nearest 0.1 percent.



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4.6.4 Microbiological tests. The product shall be microbiologically tested, as applicable, in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Microbiological Methods; Section: Examination of Frozen, Chilled, Precooked, or Prepared Foods-Official First Action. The product shall be prepared for testing by being adequately comminuted with care being taken to avoid contamination of product and moisture pickup. A representative sample shall then be removed. (For E. Coli determinations, proper dilutions shall be prepared using 25-gram samples.) The diluent shall be added to the sample of dry product and allowed to stand for 15 minutes before blending. Continue as directed by AOAC. Any of the following shall be considered a test failure:

a. Any of the five samples has a confirmed E. Coli count of greater than 10 per gram.

b. Two or more of the five samples have a confirmed E. Coli count greater than 3 per gram.

c. Any of the five samples has a total aerobic plate count greater than 75,000 per gram.

4.6.5 Fat content tests.

4.6.5.1 Soxhlet method. Fat shall be determined in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Meat and Meat Products, Section: Meat, Method; Crude Fat, using petroleum ether as solvent.

4.6.5.2 Thermal extraction method. This method shall utilize the Fat Percentage Indicator for Ground Beef, Model F-100 or F-101, manufactured by the Hobart Mfg. Co.,. Troy, OH or equal. This method shall be performed in accordance with the following procedure:

a. Initial preparation of each sample unit.

(1) Grind an approximate 8 ounce sample unit once through a constant speed grinder (electric) equipped with a plate having holes 1/8 inch in diameter to provide sufficient ground residue for testing. Thorough hand mixing of the sample unit may be substituted for the mixing accomplished by grinding.

b. Final preparation of each sample unit.

(1) Remove approximately 2 ounces of the ground meat from the constant speed grinder or after hand mixing and form it into shape of a doughnut with an outer diameter of approximately 3 inches with a hole measuring 1/2 to 1 inch in diameter in the center. Place this formed sample on the special perforated rendering pan (disc). Do not allow any portion of the sample to extend outward beyond the edge of the pan.



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(2) Weigh out accurately 56.7 grams (to nearest 0.1 gram) of the formed sample as required.

(3) Clean and dry the grinder between each test when the grinder is used for mixing of the ground beef samples.

c. Fat extraction.

(1) Follow the manufacturer's instruction for heating the patty samples, extracting (rendering) the fat and reading the amount of fat collected.

(2) Clean the rendering pan, funnel and test tube between each test.

In case of dispute regarding fat content as determined by the Thermal Extraction Method, the test method (Soxhlet) cited in 4.6.5 shall govern.

4.6.6 Seal strength test (polyethylene and plastic-foil bags). Seal strength shall be tested in accordance with the procedure specified in Method A or B of ASTM D 882. Seal specimens shall be 1/2 inch or 1 inch in width. Two nonadjacent specimens shall be taken from each of the three manufacturer's seals (bottom seal only of the polyethylene bag) of the bag to be tested. The seal strength of the processor's closure seal shall be determined in the same manner.

5. PACKAGING

5.1 Preservation. Preservation shall be level A. The product shall be filled in polyethylene bags as specified in 5.1.1. These shall then be unit packed in laminated plastic-foil bags as specified in 5.1.2.

5.1.1 Polyethylene bags. A net weight of 5 ounces of the product shall be filled into a 0.005 inch thick ± 10 percent polyethylene bag made in accordance with type I bags of PPP-B-26, except that the bag shall be made from food grade polyethylene. A net weight tolerance of minus 1/4 ounce is permitted provided that the average net weight of the bags in any lot is not less than 5 ounces (see table VII). If medium density polyethylene (0.926 to 0.940 $g/cm^3$) is used, a minimum thickness of 0.003 inch is acceptable. The film used in the bags shall not impart any undesirable odor or flavor to the product and all such films shall comply with the Food and Drug Administration Regulations for food items. The seal strength of the manufacturer's seal shall be not less than 5 pounds per linear inch (lot average) when tested in accordance with 4.6.6 with no individual unit evidencing a seal strength specimen of less than 4 pounds per linear inch. The inside dimensions of the bag shall be 5-1/2 by 3-1/2 by 8 inches ± 1/8 inch in each dimension with the open end of the bag having the 5-1/2 by 3-1/2 inch dimensions. After filling, the end panels of the bag (corresponding to the 3-1/2 inch dimension) shall be folded into form gussets, and the bag shall be closed by folding over its open end. Where required by the bulkiness of the product, the bulk of the contents within the bag shall be compressed to a depth of not more than 2-1/4 inches by applying a constant uniform pressure over the entire top surface of the product. The filled bag with



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its open top folded over to prevent spillage of the product shall be inserted to the bottom of the laminated plastic-foil bag specified in 5.1.2. Alternatively, in lieu of polyethylene bags conforming to type I of PPP-B-26, bags made in accordance with type II of PPP-B-26, with inside dimensions of 9 inches by 8 inches (± 1/8 inch in each dimension) with the open end of the bag having the 9 inch dimension, may be used. The material and seal strength shall be as specified for type I bags. When type II bags are used, they shall be formed over a baseboard on a mandrel as specified in 5.1.1.1 before filling. While the bag is positioned on the mandrel, the two bag ears shall be taped down over a 5-1/2 by 3-1/2 inch baseboard constructed in accordance with 5.1.1.2. An 8 inch long strip of 1 inch wide tape complying with type III, class 2 of PPP-T-60, shall be applied across both ears to fasten them securely to the baseboard.

5.1.1.1 Mandrel. The mandrel face shall be 5-3/8 by 3-3/8 inches and the height shall be not less than 12 inches. All corners and edges of the mandrel shall be rounded and all surfaces finished to prevent damage to the bag material.

5.1.1.2 Baseboard. The baseboard shall be constructed from unbleached and uncoated solid sulfate or sulfite paperboard not less than 0.022 inch thick. Alternatively, the baseboard may be constructed from 0.022 inch thick food grade paperboard made from recycled fibers. If paperboard made from recycled fibers is used, it shall be natural color ranging from brown to gray. The baseboard shall be 5-1/2 by 3-1/2 inches ± 1/8 inch and shall have all corners rounded to form a smooth contour.

5.1.2 Laminated plastic-foil bag. The bag shall be fabricated from a material consisting of 0.0025 inch thick ionomer or polyethylene film (± 10 percent), laminated or extrusion coated to 0.00035 inch thick ± 10 percent aluminum foil laminated on the plain side to 0.001 inch ± 20 percent thick polyester or Nylon 6. The polyester or Nylon 6 shall be bonded to the aluminum foil by hot extruding 10 pounds per ream (± 10 percent) of polyethylene between the laminae. The exterior (polyester or Nylon 6) surface of the material shall be colored overall with a color in the number ranges from 34052 to 34108 inclusive, or 20045, 20062 or 30045 of FED-STD-595. Reverse printing of the exterior layer will be acceptable and any luster contributed by the polyester or Nylon 6 lamina (external) shall not be considered as affecting the color gloss. The combined sheet material shall show no evidence of delamination before or after heat sealing. The laminated material shall be suitably formulated for food products in compliance with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder. The interior surface of the laminated material shall not impart any undesirable odor or flavor to the product. The bag shall be a flat style bag having the inside dimensions of either 6 by 10 inches ± 1/8 inch or 6 by 10-5/8 inches ± 1/8 inch and shall be heat sealed on three edges with 3/8 inch wide seals ± 1/8 inch. One side of the open end of the bag may be provided with a 1/4 inch ± 1/16 inch extended or fold-over lip to facilitate opening and filling. The laminated bag material shall have an average seal strength of not less than 6 pounds per linear inch with no individual seal strength specimen being less than 5 pounds per linear inch when tested in accordance with 4.6.6. A V-shaped or C-shaped (half round) tear notch at least 1/32 inch deep, located 1-1/4 to 1-1/2 inch from the top edge of the bag shall be made in both side seals. The distance between the inside edge of the tear notch and the inside edge of the seal shall be at least 3/16 inch. The unit pack shall be closed as specified in 5.1.2.1.



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5.1.2.1 Closure of filled laminated plastic-foil bag. The filled laminated bag shall be evacuated and closed by heat sealing the open end in a vacuum of not less than 28 inches of mercury (an absolute pressure of not more than 6 kPa) as registered on a vacuum gauge on the closing machine. After a storage period of 24 hours or more following the unit pack closing operation, an examination shall be made of each unit pack to determine compliance with the vacuum requirements (see 4.5.4.1 for procedure). The closure seal shall be not less than 1/4 inch nor more than 1/2 inch wide and shall have an average seal strength of not less than 6 pounds per linear inch with no individual seal strength specimen less than 5 pounds per linear inch when tested in accordance with 4.6.6. Alternatively, an optional double seal may be applied to close the bags. The double closure seal shall be at least 5/16 inch wide and shall have an average seal strength of 6 pounds per linear inch and no individual specimen shall have a strength of less than 5 pounds per linear inch when tested in accordance with 4.6.6. If thermal impulse sealing is used for the closure seal, a seal width of not less than 1/8 inch is required. The closure seal shall be above the tear notches. The unit pack shall be labeled in accordance with 5.4, subsequent to the closing operation. Unit packs which are improperly sealed or fail to maintain vacuum after sealing may be resealed or the contents may be placed in a new laminated plastic-foil bag and resealed provided that the product is repackaged within 24 hours and the product is adequately protected from moisture and oxygen (see 3.4).

5.2 Packing.

5.2.1 Level C (for shipment to the ration assembler). Forty-five filled and sealed unit packs, preserved as specified in 5.1, shall be packed in a fiberboard box with liner, complying with the Uniform Freight Classification or National Motor Freight Classification, as applicable. In addition, top and bottom pads constructed of the same material as the shipping container shall be used in packing. The unit packs shall be packed flat or nested in three layers of fifteen unit packs each. Each unit pack shall be packed with its excess package material maintained in the flat (unfolded) state. Dunnage material not less than 1/8 inch thick, conforming to type II, class B of PPP-C-843, or, alternatively, dunnage material not less than 1/16 inch thick, conforming to PPP-C-1797, shall be placed between each of the three layers of unit packs and in the bottom and top of the cases to minimize movement of each individual packet. The length and width of the pieces of dunnage material shall be not less than the length and width of the shipping container.

5.3 Unit loads. When specified (see 6.1), the product packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.4 Labeling and marking.

5.4.1 Unit pack, (food packets). Each unit pack shall have labeling that complies with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder. The label shall be pressure-sensitive and be colored a dull non-reflective color other than white, and all printing thereon shall be black, boldface, legible type with indelible ink. The label shall be applied to one face of the bag with sufficient pressure to assure its adherence to the bag. The labels shall contain the following information:



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CAUTION - INNER BAG IS OPEN ON ONE SIDE

REMOVE IT CAREFULLY HOLDING OUTER
BAG UPRIGHT TO AVOID SPILLING CONTENTS

NAME OF PRODUCT

TEAR BAG OPEN AT NOTCHES. ADD 1/2 CANTEEN CUP HOT OR COLD WATER TO FOOD IN BAG. LET SOAK ABOUT 5 MINUTES. ADD MORE WATER IF DESIRED. MAY BE EATEN DRY.

NET WT. 5 AVD. OZ. (142 g)

NAME AND ADDRESS OF PRODUCER

USDA INSPECTION MARKING

5.4.1.1 Alternative printing on unit packs (food packets). Alternatively, the above information may be printed directly on the bag with indelible black ink. When printing is applied directly to the bag it shall be located in an area that will permit the printing to be legible after vacuum sealing.

5.4.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1).
(c) Type and class of unit load, when unit loading is required (see 5.3).

6.2 Award of contract. Award of contract for the product specified in this document will be limited to plants known to maintain the required sanitation conditions of 3.7 and 4.3.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The first article should consist of twelve bags. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.



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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-H-43224G Hash, Beef, Dehydrated, Cooked

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

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